SESSION 7

UML Sequence and Activity Diagrams

AI Career Guidance & Skill Recommendation System

Project	AI Career Guidance & Skill Recommendation System
Deliverable	Sequence and Activity Diagram
Industry Mapping	Interaction and Workflow Modeling
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Repository	[Add GitHub repository link]

Topics Covered in This Session

- Sequence diagram design
- Message flow representation
- Workflow modelling using activity diagrams

1. Sequence Diagrams

Sequence diagrams show **who talks to whom and in what order**. Vertical dashed lines are lifelines. Solid arrows are calls, dashed arrows are returns, and a loop arrow on one lifeline is a self call (work done inside the same component).

Participant	Role
Student	Actor who uploads a resume or asks a question
React UI	Client application in the browser
Express API	Controllers and services on the Node.js server
Resume Parser	Text extraction module for PDF and DOCX files
Claude API	External AI service
MongoDB	Database storing users, resumes, careers, results

1.1 Sequence Diagram: Resume Analysis and Career Recommendation

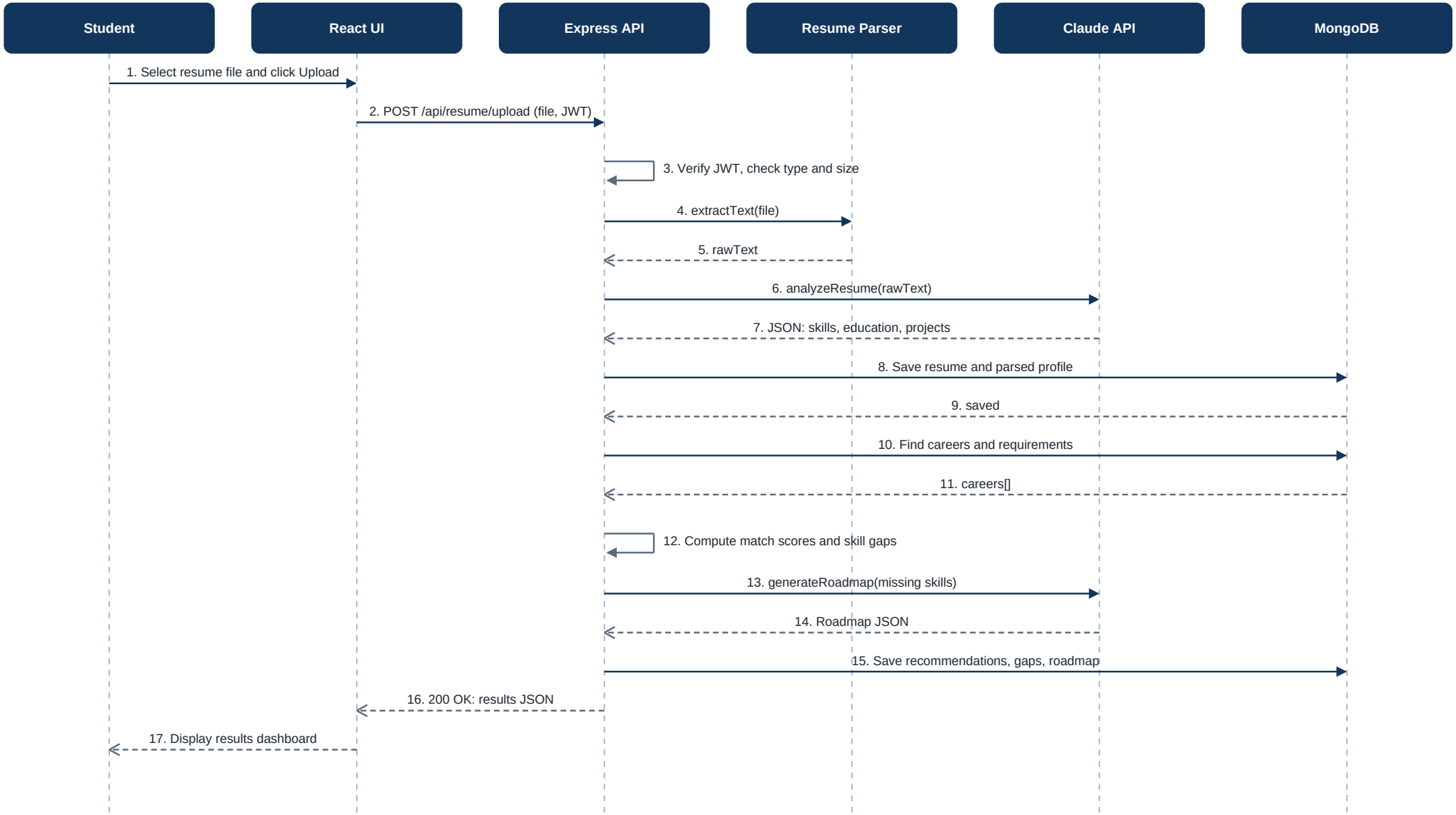

Figure 1: Resume upload, AI analysis, recommendation and roadmap generation

1.2 Sequence Diagram: AI Career Chatbot

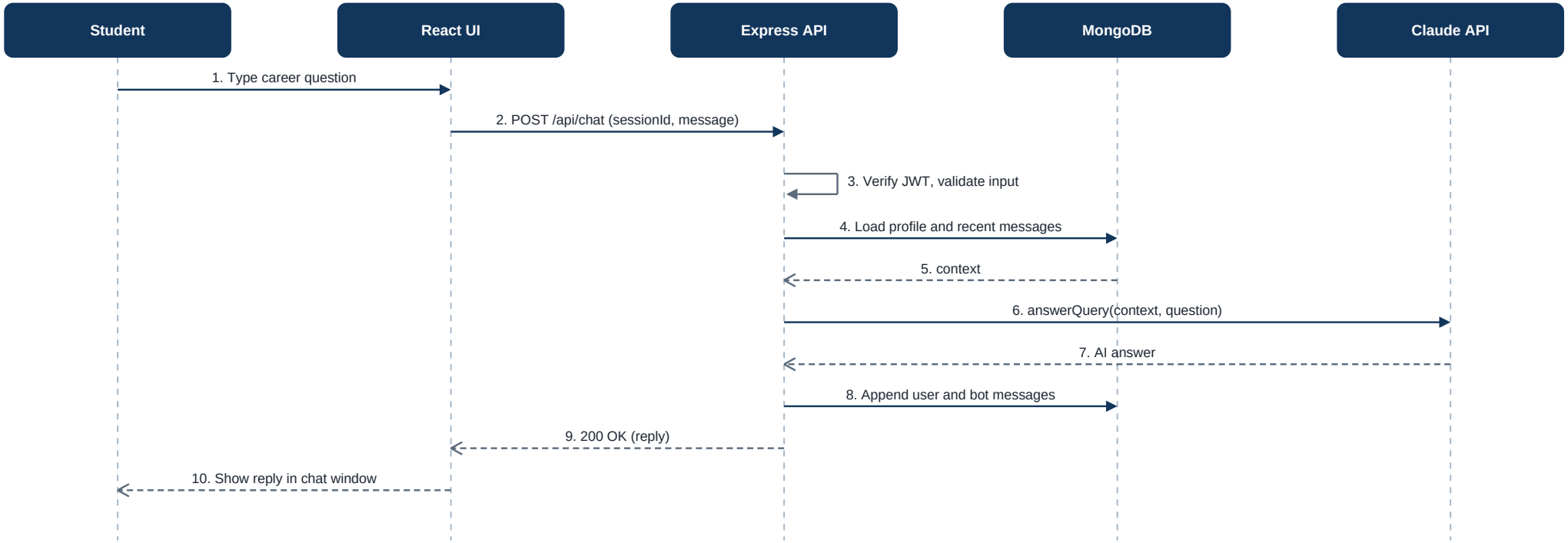

Figure 2: Chatbot question and answer flow

Message flow notes

- Every request carries a JWT; the API verifies it before any work (self call at the start of each flow).
- The Claude API key stays on the server; the browser never calls Claude directly.
- Profile and recent messages are loaded from MongoDB and added to the prompt, so answers are personalised.
- On Claude failure the API returns an error response and the UI offers a retry (exception path not drawn for clarity).

2. Activity Diagram

The activity diagram shows the end-to-end student workflow using three swimlanes. Diamonds are decisions; amber activities are performed by Claude AI.

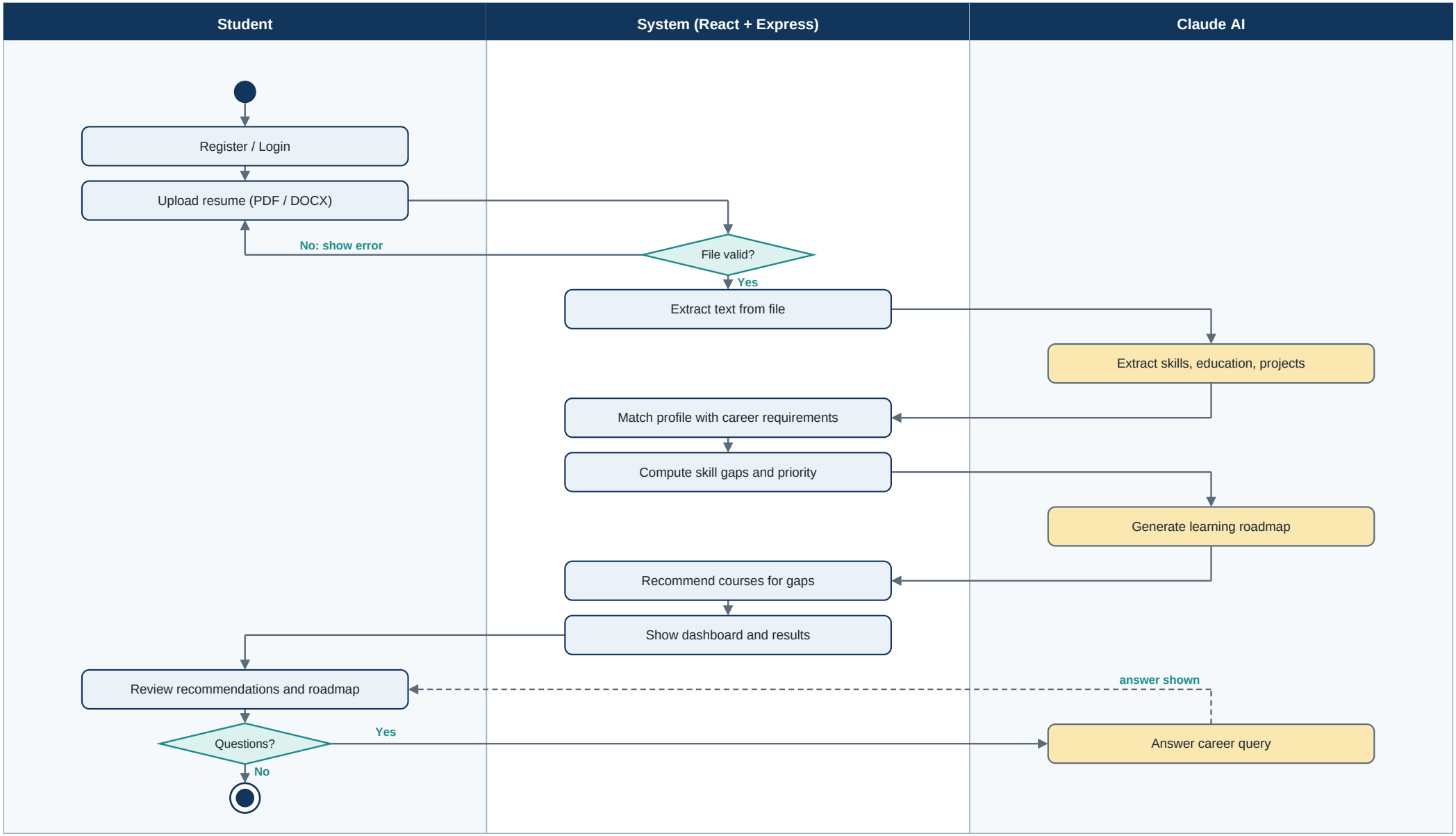

Figure 3: Activity diagram of the student workflow with swimlanes

3. Diagram to Requirement Mapping

Diagram element	Related requirement
File valid? decision	FR-06 (accept only PDF/DOCX up to 5 MB)
Extract skills (Claude)	FR-08 (structured extraction using AI)
Match profile with career requirements	FR-11 to FR-13 (weighted match score)
Compute skill gaps	FR-15, FR-16
Generate learning roadmap	FR-17
Recommend courses	FR-19
Questions? loop to chatbot	FR-26, FR-27